Abhijit Chowdhary

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About Me

I am a computational mathematician and my research interests lie in wielding math to design algorithms for real-world problems. Largely, I practice that in the fields of uncertainty quantification and optimization. My work starts from a simple observation, that large inverse and learning problems are rarely hard everywhere. I look for the few directions the data actually informs, or the blocks of a problem that can be solved exactly, and build algorithms around them. Currently, I plan to bring these ideas to inverse problems with learned priors, which often act as black boxes that hide this structure.

Education

North Carolina State University, PhD in Applied Mathematics – Raleigh, NC — Aug 2020 – May 2025

- Recipient of the Siewert Fellowship

New York University, BA in Mathematics, Computer Science – New York, NY — May 2016 – May 2020

- Minor in Classics

Research Appointments

Postdoctoral Visiting Scholar, ICERM, Brown University – Providence, RI — Jan 2026 – Apr 2026

- Began the collaboration behind the MarCo sampler, in *Exploiting Exact Conditionals Improves Conditioning*.
- Began collaborations on several other projects, now in preparation.

Postdoctoral Researcher, Tufts University – Medford, MA — June 2025 – Present

- Working with Elizabeth Newman in structure-exploiting SciML and uncertainty quantification.
- Developed VPBoost, in *Boost Like a (Var)Pro: Trust-Region Gradient Boosting via Variable Projection*.
- Funded by Elizabeth Newman's 2025 AFOSR Young Investigator Program (YIP) Award (FA9550-26-1-B049).

Givens Associate, Argonne National Laboratory – Lemont, IL — May 2024 – Aug 2024

Givens Associate, Argonne National Laboratory – Lemont, IL — May 2023 – Aug 2023

- Became a core maintainer of the PyOED project and published a peer reviewed paper in TOMS.
- Began research in robust optimal experimental design, later published in SISC.
- Under the mentorship of Ahmed Attia.

Research Assistant, North Carolina State University – Raleigh, NC — June 2021 – Aug 2024

- Published novel research in scalable UQ for infinite-dimensional Bayesian inverse problems.
- Advised by Alen Alexanderian and funded by NSF Grant DMS-2111044.

Undergraduate Research Assistant, Ohio University – Athens, OH — May 2019 – Aug 2019

- Published novel research in the theory of pattern formation, for the 2D Swift-Hohenberg Equation.

Teaching

Instructor, Tufts University – Medford, MA

- Will teach Mathematical Modeling in Spring 2027.
- Teaching Mathematical Aspects of Data Analysis in Fall 2026.
- Led recitations for Linear Algebra in Fall 2025.

Teaching Assistant, North Carolina State University – Raleigh, NC

- Led recitations for Calculus I, II, and the honors variants for 5 semesters.

Preprints

Exploiting Exact Conditionals Improves Conditioning: Provably Fast Mixing Time Bounds By Sampling from the Marginal Aug 2026
Abhijit Chowdhary, Federica Milinanni, Julianne Chung, Elizabeth Newman
https://doi.org/10.48550/arXiv.2608.27884 (arXiv preprint, submitted to SIAM JUQ)

Boost Like a (Var)Pro: Trust-Region Gradient Boosting via Variable Projection Mar 2026
Abhijit Chowdhary, Elizabeth Newman, Deepanshu Verma
https://doi.org/10.48550/arXiv.2603.23658 (arXiv preprint, submitted to JMLR)

Publications

Robust Optimal Experimental Design of Infinite-Dimensional Bayesian Nonlinear Inverse Problems Apr 2026
Abhijit Chowdhary, Ahmed Attia, Alen Alexanderian
https://doi.org/10.1137/24M1693921 (SIAM Journal on Scientific Computing)

Weak Diffusive Stability of Roll Solutions at the Zigzag Boundary May 2025
Abhijit Chowdhary, Mason Haberle, William Ofori-Atta, Qiliang Wu
https://doi.org/10.1137/24M1639683 (SIAM Journal on Mathematical Analysis)

Scalable Uncertainty Quantification for Infinite-Dimensional Bayesian Inverse Problems Feb 2025
Abhijit Chowdhary
www.lib.ncsu.edu/resolver/1840.20/45106 (PhD Thesis, NC State University)

PyOED: An Extensible Suite for Data Assimilation and Model-Constrained Optimal Design of Experiments June 2024
Abhijit Chowdhary, Shady E. Ahmed, Ahmed Attia
https://dl.acm.org/doi/10.1145/3653071 (ACM Transactions on Mathematical Software)

Sensitivity Analysis of the Information Gain in Infinite-Dimensional Bayesian Linear Inverse Problems May 2024
Abhijit Chowdhary, Shanyin Tong, Georg Stadler, Alen Alexanderian
https://doi.org/10.1615/Int.J.UncertaintyQuantification.2024051416 (IJUQ)

Software

PyOED May 2023 – May 2025

- Core maintainer of PyOED, an open source research software for model-constrained optimal design.
- Designed to extend and integrate existing codes to rapidly test new developments.
- Used in multiple research groups worldwide.

Talks

Randomized Strategies for Variable Projection in Hierarchical Inverse Problems Nov 2026

- SIAM IS 2026, Salt Lake City, UT (Nov 2026).

VPBoost: Variable Projection for Gradient Boosting and VarProNets Dec 2025 – Present

- SIAM MDS 2026, Salt Lake City, UT (Nov 2026).
- WCCM–ECCOMAS 2026, Munich, Germany (July 2026).
- CMO Integrating Data- and Physics-Driven Methods for Decision Making under Uncertainty, Oaxaca, MX (June 2026).
- SIAM UQ 2026, Minneapolis, MN (Mar 2026).
- ICERM, Providence, RI (Feb 2026).
- CBMS Conference, University of Houston, Houston, TX (Dec 2025).

Robust Optimal Experimental Design for Large-Scale Bayesian Inverse Problems May 2023 – Mar 2025

- SIAM CSE 2025, Fort Worth, TX (Mar 2025).
- Applied Math Graduate Student Seminar, NC State (Nov 2024).
- SIAM MDS 2024, Atlanta, GA (Oct 2024).
- SASSy Seminar, Argonne National Laboratory (May 2023).

PyOED for Bayesian Optimal Experimental Design Feb 2024 – Aug 2024

- SASSy Seminar, Argonne National Laboratory (Aug 2024).
- SIAM UQ 2024, Trieste, Italy (Feb 2024).

Sensitivity Analysis and Scalable Optimal Design for Bayesian Inference Apr 2022 – Aug 2023

- Applied Math Graduate Student Seminar, NC State (Aug 2023).
- SIAM CSE 2023, Amsterdam, Netherlands (Mar 2023).
- Applied Math Graduate Student Seminar, NC State (Apr and Sept 2022).

Service

Minisymposium Co-Organizer

- *Structure-Exploiting Techniques for UQ and ML*, SIAM UQ 2026, Minneapolis, MN (Mar 2026).
- *Lightweight Learning: Exploiting Structure for Reliable and Efficient ML*, SIAM MDS 2026, Salt Lake City, UT (Nov 2026).

Group Leader, VERSEIM-REU June 2026 – Aug 2026

- Co-advised three undergraduate researchers with Elizabeth Newman on a new training scheme for generative models.
- The students will present their poster at the 2027 Joint Mathematics Meetings.

Co-Organizer, Postdoctoral and Graduate Student Seminar, ICERM Jan 2026 – Apr 2026

Lead Organizer, Applied Math Graduate Student Seminar (AMGSS), NCSU Jan 2022 – May 2025

Referee

- Journal of Open Source Software (JOSS)
- SIAM Journal on Matrix Analysis and Applications (SIMAX)
- SIAM Journal on Uncertainty Quantification (JUQ)